

# Milestone 4 DONE

EML Viewer Chrome Extension · Attachment Previews + Polish + Team Handoff

## In plain English

The last pieces. Attachments are now **clickable**: click one and it opens in its own tab — **images show as images, PDFs open in a real PDF viewer**, and anything else gives you a **Download** button. We also added the finishing touches (a loading spinner, an "Open another file" button, friendly error messages) and packaged everything so it can be **handed to the team and installed**.

This completes the project. The extension does what it set out to do: open `.eml` files safely and privately, with no internet access.

## How attachment opening works

Clicking an attachment in the list opens a new tab that loads the file *from the local database* (never the network) and decides how to show it:

| File type                 | What happens                                                                            |
|---------------------------|-----------------------------------------------------------------------------------------|
| Images (PNG, JPG, GIF...) | Shown inline, scaled to fit, with a Download button.                                    |
| PDF                       | Rendered by <b>Chrome's built-in PDF viewer</b> inside the tab, plus a Download button. |
| Everything else           | Filename + size + a <b>Download</b> button (no preview).                                |

**A design decision worth noting:** we originally planned to show PDFs inside the locked sandbox, but a fully locked sandbox blocks the PDF plugin — it would show nothing. Since attachments are *your own team's files* (trusted, unlike email body HTML), we show the PDF in a normal frame on the extension's own page, which lets Chrome's native viewer work. The file is loaded as a temporary in-memory link that's released when you close the tab.

## Polish added

- **Loading spinner** while a file is parsed and saved.
- **"Open another file"** button in the reader tab, so you don't have to go back to the icon.

- **Friendly error messages** for: wrong file type, file too big (50 MB cap), corrupt/unparseable email, an attachment or email that's no longer available, and emails with no readable body.

## What we built this milestone

| File               | Role                                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| preview.js         | Decides image / pdf / download for a given file type (unit-tested).                                         |
| attachment_page    | The preview tab: header with name/size/Download, then inline image or native PDF or a "no preview" message. |
| viewer_page        | Attachment list items are now buttons that open the preview tab; added the toolbar.                         |
| HANDOFF.md         | Build + install guide (load-unpacked and enterprise force-install).                                         |
| scripts/package.sh | <code>npm run package</code> → builds and zips the extension for distribution.                              |

## Proof it works (automated)

- **26 unit + integration tests pass** (up from 20). New: the preview router, plus real parsing of a **PDF attachment**, a **nested forwarded email** ( `message/rfc822` ), and an **empty-body** email.
- **Packaging verified:** `npm run package` produces `platformeq- eml-viewer.zip` with a clean layout (manifest at the root, one icons folder, no duplicates).

### ✅ Your final check in Chrome (~4 minutes)

1. `chrome://extensions` → **reload** the EML Viewer.
2. Drop `samples/with-pdf.eml` → click **report.pdf** in the list → it opens in a new tab and the **PDF renders**. The **Download** button saves it.
3. Drop `samples/inline-image.eml` → click **note.txt** → new tab shows name/size and a **Download** button (no preview, as expected).
4. Drop `samples/nested.eml` → **forwarded.eml** is listed and downloads.
5. Drop `samples/empty-body.eml` → opens with a tidy "no readable body" note (no crash).
6. In a reader tab, click **"Open another file"** → the import page opens.

If those all behave, the project is complete and ready to package for the team.



## Project complete — what you can do now

- **Use it:** load dist/ unpacked (see HANDOFF.md).
- **Share it:** `npm run package` → send `platformeq-eml-viewer.zip`, or have IT force-install it (steps in HANDOFF.md).

## Out of scope (possible follow-ups)

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- Reply / forward / search across emails; a persistent mailbox history.
- `.msg` (Outlook binary) support.
- Dark mode / theming.
- Optimizing very large inline images (currently stored as text-encoded data URLs).